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Real-Time Food Delivery Analytics and Customer Satisfaction Prediction

A CAPSTONE PROJECT REPORT

CSA1610-Dataware Housing and Data Mining

to the award of the degree of

BACHELOR OF ENGINEERING

IN

B. Tech AI&ML

Submitted by

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BONAFIDE CERTIFICATE

This is to certify that the Capstone Project entitled “**Real-Time Food Delivery Analytics and Customer Satisfaction Prediction**” has been carried out by **G. Hima Bindu(192511377)** under the supervision of **Dr. K. V. Kanimozhi** and is submitted in partial fulfilment of the requirements for the current semester of the **B. Tech- Computer Science and Engineering(AI&ML)** program at Saveetha Institute of Medical and Technical Sciences, Chennai.

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DECLARATION

I, G.HimaBindu(192511377) of the B.Tech-ComputerScience and Engineering(AI&ML) , SIMATS Engineering, Saveetha Institute of Medical and Technical Sciences, Chennai, hereby declare that the Capstone Project Work entitled “Diabetes Risk Prediction Application” is the result of our own bonafide efforts. To the best of our knowledge, the work presented herein is original, accurate, and has been carried out in accordance with the principles of engineering ethics and academic integrity. All sources, references, data, and other materials used in the project have been appropriately acknowledged. We further declare that the data, results, analysis, and findings presented in this report have not been fabricated, falsified, or manipulated. Any external tools, software, datasets, computational resources, or AI-assisted tools used during the project have been appropriately disclosed. We confirm that all applicable ethical, academic, institutional, and professional requirements have been duly followed throughout the planning, execution, analysis, and documentation of the project.

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Individual Contribution Statement

Student Name / Register No.	Specific Responsibilities	Design & Development Contribution	Testing & Analysis Contribution	Report Contribution	Approx. Contribution (%)
G.HimaBindu (192511377)	Module 1 – Data Collection	Data collection & validation	Input testing	Introduction & Module 1	[<u>34</u> %]
Total					100%

ABSTRACT

Real-time food delivery platforms face significant engineering challenges in managing fluctuating customer demand, delivery delays, restaurant preparation times, rider availability, and customer expectations. These factors directly influence delivery performance and customer satisfaction. Therefore, an intelligent analytics system is needed to process delivery data in real time and predict customer satisfaction accurately, enabling food delivery platforms to improve service quality and operational efficiency.

The proposed Real-Time Food Delivery Analytics and Customer Satisfaction Prediction system uses data analytics and machine learning techniques to analyze important factors such as order time, preparation duration, delivery distance, traffic conditions, delivery time, order value, restaurant performance, and customer ratings. The methodology involves collecting and preprocessing real-time food delivery data, performing exploratory data analysis, extracting relevant features, and developing a machine learning prediction model. The system can classify or predict customer satisfaction based on delivery-related parameters. A real-time analytics dashboard can also be used to monitor delivery performance, identify delays, and provide useful insights to restaurants and delivery managers.

The major results are expected to demonstrate that delivery time, food preparation time, traffic conditions, and order accuracy are among the most influential factors affecting customer satisfaction. The prediction model can provide reliable satisfaction predictions, while real-time analytics can help identify potential delivery problems before they negatively affect customers. Performance evaluation can be carried out using accuracy, precision, recall, F1-score, and other suitable metrics.

The principal conclusion is that integrating real-time analytics with machine learning provides an effective approach for predicting customer satisfaction and improving food delivery operations. The proposed system can support faster decision-making, reduce delivery delays, improve resource allocation, and enhance the overall customer experience. Thus, the system provides a practical engineering solution for developing more efficient, responsive, and customer-oriented food delivery services.

Keywords: Real-Time Analytics, Food Delivery, Customer Satisfaction, Machine Learning, Predictive Analytics, Delivery Optimization

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LIST OF ABBREVIATIONS / SYMBOLS

Symbol	Description	Unit

CHAPTER 1

INTRODUCTION AND ENGINEERING PROBLEM

1.1 Background

Food delivery services have become an important part of modern life because customers can easily order food through online platforms. These platforms generate a large amount of information related to customer orders, restaurants, delivery time, payment, ratings, and feedback.

Analyzing food delivery data can help understand customer behaviour and identify the factors that influence customer satisfaction. However, raw transaction data may be large, unorganized, and difficult to analyze directly.

Data warehousing provides a structured method to collect, organize, and store delivery transaction information. This structured data can then be used for further analysis and prediction.

Machine Learning and data mining techniques provide the capability to analyze multiple delivery and customer-related parameters and identify patterns associated with customer satisfaction. The proposed **Real-Time Food Delivery Analytics and Customer Satisfaction Prediction** system uses delivery transaction information and customer-related data to analyze patterns and predict customer satisfaction.

The project processes parameters including delivery time, order value, customer rating, food quality, discounts, order accuracy, restaurant information, and other relevant data.

The complete project is divided into three major modules:

- **Module 1 – Delivery Transaction Data Warehousing**
- **Module 2 – Satisfaction Prediction and Pattern Mining**
- **Module 3 – Customer Satisfaction Prediction**

Module 1 is the starting point of this complete pipeline.

1.2 Need for the Project

Customer satisfaction is very important in food delivery services. Factors such as late delivery, incorrect orders, poor food quality, and service issues may affect customer satisfaction.

The project is required because:

1. Food delivery platforms generate large amounts of transaction data.
2. Raw transaction data may be difficult to analyse directly.
3. Customer satisfaction depends on multiple delivery-related factors.
4. Manual analysis of large amounts of transaction data can be time-consuming.
5. Data warehousing helps organize delivery transaction information.
6. Data mining techniques can identify important customer satisfaction patterns.
7. Machine-learning techniques can predict customer satisfaction.
8. A structured data collection and storage process is necessary before further analysis.

Therefore, Module 1 is important because it establishes a reliable data storage and organization stage for the complete customer satisfaction prediction system.

1.3 Problem Statement

To design and develop a structured **Delivery Transaction Data Warehousing Module** for a Real-Time Food Delivery Analytics and Customer Satisfaction Prediction system that collects, organizes, and stores relevant food delivery transaction data and provides reliable information for subsequent pattern mining and customer satisfaction prediction.

The broader project problem identified is:

To develop an intelligent and user-friendly system that can analyse food delivery transaction data, identify customer satisfaction patterns, and predict customer satisfaction using data analytics and Machine Learning techniques, thereby supporting improved food delivery services and customer experience.

1.4 Project Aim

The aim of **Module 1** is to:

Develop a structured and reliable delivery transaction data warehousing system that collects, organizes, and stores relevant food delivery information before forwarding the data to the satisfaction analysis and prediction stages.

1.5 Project Objectives

The objectives of **Module 1** are:

1. To identify the relevant delivery transaction parameters required for customer satisfaction analysis.
2. To collect food delivery transaction information.
3. To organize transaction data into a structured format.
4. To store customer and delivery-related information.
5. To identify missing or inconsistent transaction data.
6. To prepare structured delivery information for further analysis.
7. To provide organized data to the pattern mining module.
8. To create a scalable data warehousing mechanism that can support additional delivery parameters in the future.

1.6 Scope

Included

The scope of Module 1 includes:

- Collection of food delivery transaction information.
- Customer order details.
- Delivery time information.
- Restaurant information.
- Order value.
- Customer rating.
- Payment information.
- Discount information.
- Delivery status.

- Basic data validation.
- Identification of missing or invalid values.
- Storage of structured transaction data.
- Preparation of data for Module 2.

Excluded

The following are outside the primary scope of Module 1:

- Customer satisfaction prediction.
- Pattern mining algorithm execution.
- Machine-learning model training.
- Final satisfaction prediction.
- Real-time delivery tracking.
- Personalized customer recommendations.
- Direct management of food delivery operations.

Assumptions

The module assumes that:

- Food delivery transaction information is available.
- The collected data contains relevant delivery parameters.
- Customer ratings and feedback information are available where required.
- The transaction data is suitable for further analysis.
- The system is used for analysing delivery performance and customer satisfaction.

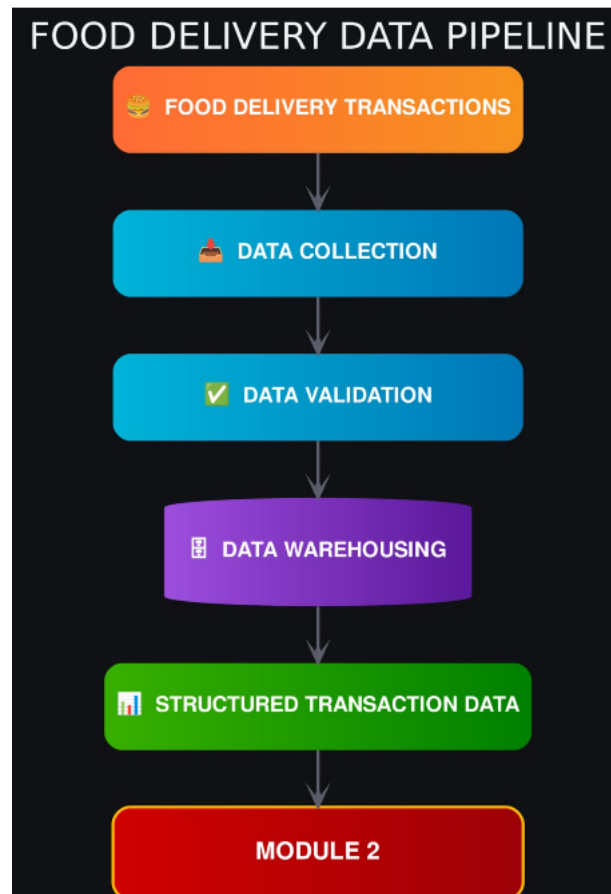
Boundary Conditions

Module 1 ends after the delivery transaction information has been collected, organized, stored, and prepared for forwarding to the pattern mining and satisfaction prediction stage.

1.7 Expected Engineering Outcome

The expected outcome of Module 1 is a functional **Delivery Transaction Data Warehousing Module** capable of:

Fig No:1.1 Food Delivery Data Pipeline



CHAPTER 2

LITERATURE REVIEW AND EXISTING SOLUTIONS

2.1 Review Method

Relevant research on food delivery analytics, customer satisfaction prediction, data warehousing, data mining, and machine-learning-based customer analysis was reviewed to understand existing approaches, commonly used datasets, prediction techniques, and important factors affecting customer satisfaction.

The review considered:

- Food delivery analytics.
- Customer satisfaction prediction.
- Delivery transaction data warehousing.
- Pattern mining techniques.
- Machine-learning-based prediction.
- Customer ratings and feedback analysis.
- Delivery time analysis.
- Data preprocessing and feature selection.
- Real-time data analytics.

The literature and existing solutions identified include approaches involving data warehousing, classification algorithms, pattern mining, customer feedback analysis, decision trees, Random Forest, Logistic Regression, and other machine-learning techniques.

2.2 Review of Existing Work

Study 1 – Food Delivery Customer Satisfaction Analysis

Previous research on online food delivery services has examined factors affecting customer satisfaction, including delivery time, food quality, service quality, order accuracy, and price.

Finding: Multiple service-related factors can influence the overall satisfaction of food delivery customers.

Limitation: Customer satisfaction may vary depending on location, customer preferences, and service conditions.

Study 2 – Food Delivery Transaction Data Analysis

Research on food delivery platforms has used transaction data to analyze customer orders, delivery performance, restaurant services, and customer behaviour.

Finding: Structured transaction data can provide useful insights into customer behaviour and delivery performance.

Limitation: Raw transaction data may contain missing, inconsistent, or unstructured information.

Study 3 – Machine Learning for Customer Satisfaction Prediction

Machine-learning approaches have been used to predict customer satisfaction based on customer ratings, service quality, delivery time, and transaction information.

Finding: Machine-learning techniques can identify relationships between multiple service parameters and customer satisfaction.

Limitation: Prediction performance depends on the quality and availability of historical data.

Study 4 – Pattern Mining in Customer Data

Data mining techniques have been used to identify patterns in customer transactions and service-related data.

Finding: Pattern mining can identify relationships between customer behaviour, delivery performance, and satisfaction.

Limitation: Large and complex datasets may require effective preprocessing before pattern analysis.

Study 5 – Real-Time Food Delivery Analytics

Existing food delivery systems generate large amounts of real-time transaction and delivery information. Analytics techniques can be used to monitor and analyse delivery-related performance.

Finding: Real-time analytics can support faster identification of delivery trends and customer satisfaction factors.

Limitation: Real-time data processing requires properly organized and structured data.

2.2 Comparative Analysis

Table 2.1 – Comparative Analysis of Existing Approaches

Existing Approach	Advantage	Limitation	Relevance
Manual customer feedback analysis	Easy to understand customer opinions	Time-consuming for large data	Provides background
Traditional data storage	Stores transaction information	Difficult to analyse large datasets efficiently	Alternative approach
Data Warehousing	Organizes large transaction data	Requires proper data organization	Selected approach for Module 1
Pattern Mining	Identifies hidden patterns	Depends on data quality	Selected approach for Module 2
Machine Learning	Can predict customer satisfaction	Requires training data	Selected approach for Module 3
Real-Time Analytics	Provides faster insights	Requires continuous data processing	Important for complete system
Proposed Delivery Transaction Data Warehousing Module	Structured transaction data	Depends on available transaction information	Initial project stage

2.3 Research / Engineering Gap

The reviewed approaches demonstrate that data analytics and machine learning can be useful for analysing food delivery performance and predicting customer satisfaction.

However, an effective customer satisfaction prediction system requires properly organized and reliable transaction data.

A major engineering gap addressed by **Module 1** is the structured collection and storage of food delivery transaction information before the data analysis and prediction stages.

If transaction data is:

- Missing,
- Incorrect,
- Inconsistent, or
- Inappropriately formatted,

the subsequent pattern mining and customer satisfaction prediction process may be affected.

Similarly, customer satisfaction patterns may not be clearly identified without proper data analysis.

Therefore, the proposed project emphasizes:

- Structured transaction data warehousing.
- Pattern mining for satisfaction analysis.
- Machine-learning-based customer satisfaction prediction.

2.4 Proposed Contribution

The proposed contribution of the complete project is the development of a **Real-Time Food Delivery Analytics and Customer Satisfaction Prediction System** consisting of three modules.

Module 1 – Delivery Transaction Data Warehousing

The proposed contribution of Module 1 is the development of a structured data layer that:

- Collects food delivery transaction information.

- Stores customer order details.
- Organizes delivery-related data.
- Identifies missing or inconsistent data.
- Produces structured transaction information.
- Transfers organized data to Module 2.

Module 2 – Satisfaction Prediction and Pattern Mining

The proposed contribution of Module 2 is to:

- Analyze structured food delivery data.
- Identify customer satisfaction patterns.
- Analyze delivery performance.
- Identify important factors affecting satisfaction.
- Discover relationships between delivery information and customer ratings.
- Transfer analyzed information to Module 3.

Module 3 – Customer Satisfaction Prediction

The proposed contribution of Module 3 is to:

- Use processed delivery data for prediction.
- Apply machine-learning techniques.
- Predict customer satisfaction levels.
- Classify customers based on satisfaction.
- Generate the final customer satisfaction result.

Complete Project Contribution

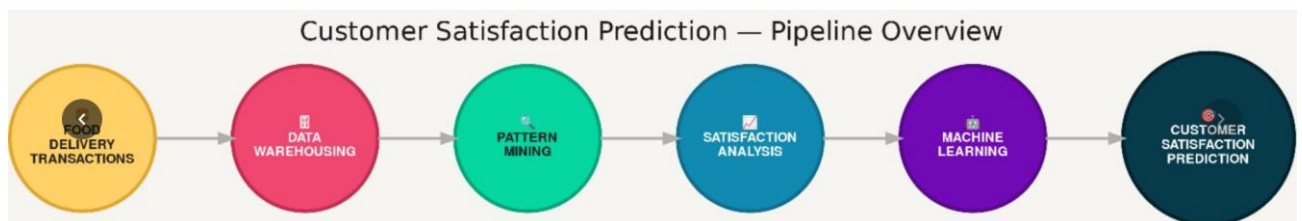


Fig No 2.1: Customer Satisfaction Prediction

CHAPTER 3

ENGINEERING DESIGN & TRADE-OFFS, SUSTAINABILITY, ETHICS, SAFETY, RISK & SECURITY

Table 3.1: Functional & Non-Functional Requirements

Requirement	Type	Target
Accept delivery transaction data	Functional	Required transaction data accepted
Store transaction data	Functional	Structured data warehousing
Validate data	Functional	Invalid data identified
Detect missing values	Functional	Missing fields identified
Organize data	Functional	Structured transaction data generated
Pattern mining	Functional	Important satisfaction patterns identified
Satisfaction prediction	Functional	Customer satisfaction predicted
Easy interface	Non-functional	Simple user interaction
Fast processing	Non-functional	Quick analytics and prediction
Scalability	Non-functional	Additional data parameters supported

3.2 Constraints & Applicable Standards

Technical Constraints

- The system depends on available food delivery transaction data.
- Incorrect or incomplete transaction data can affect analysis.
- Large amounts of transaction data may require efficient storage.
- The data format must remain compatible between all three modules.
- Machine-learning prediction depends on the quality of historical data.
- Real-time processing may require additional computational resources.

Ethical Constraints

- Customer information must be handled responsibly.
- Personal customer information should not be unnecessarily exposed.
- Customer satisfaction predictions should be used responsibly.
- The system should avoid unfair analysis based on irrelevant personal information.
- Customer data should be protected during storage and processing.

3.3 Design Specifications

Table 3.3

Parameter	Requirement	Priority
Transaction parameters	Order ID, Delivery Time, Rating, Order Value	High
Data warehousing	Required	High
Data format	Structured	High
Pattern mining	Required	High
Satisfaction prediction	Required	High
Interface	Simple	High
Data transfer	Compatible with all modules	High
Scalability	Additional parameters possible	Medium
Real-time analytics	Faster processing	Medium

3.4 Alternative Solutions & Evaluation

Alternative 1 – Manual Transaction Analysis

The developer manually analyses food delivery transaction data and customer ratings.

Advantage: Easy to understand and implement.

Alternative 2 – Dataset-Based Analysis

The system receives food delivery information from an existing dataset and performs analysis.

Advantage: Useful for testing, training, and developing the prediction system.

Alternative 3 – Integrated Food Delivery Analytics System

The system collects transaction data, organizes it in a data warehouse, identifies patterns, and automatically predicts customer satisfaction.

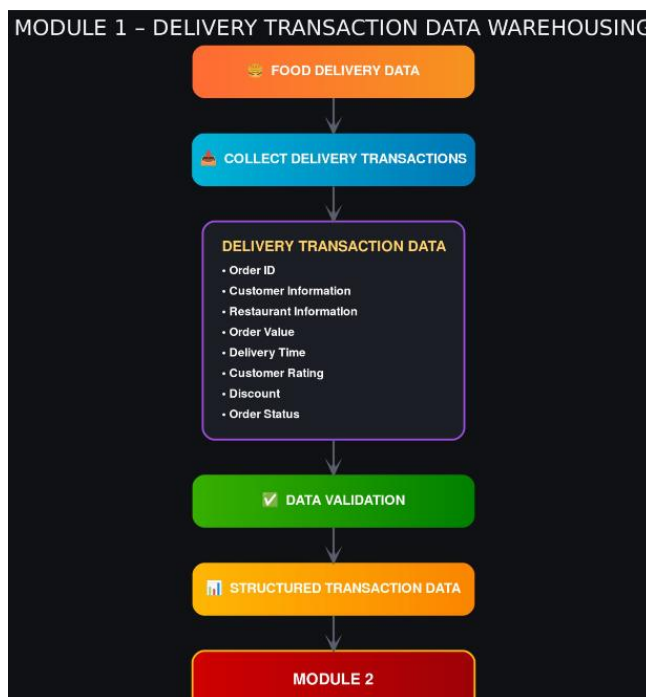
Advantage: Provides better integration with the complete analytics and prediction pipeline.

Alternative Solutions Comparison

Criterion	Manual Analysis	Dataset-Based Analysis	Integrated System
Simplicity	High	Medium	High
User interaction	High	Low	High
Data integration	Low	Medium	High
Scalability	Low	Medium	High
Pattern analysis	Low	High	High
Satisfaction prediction	Low	Medium	High
Real-time capability	Low	Low	High

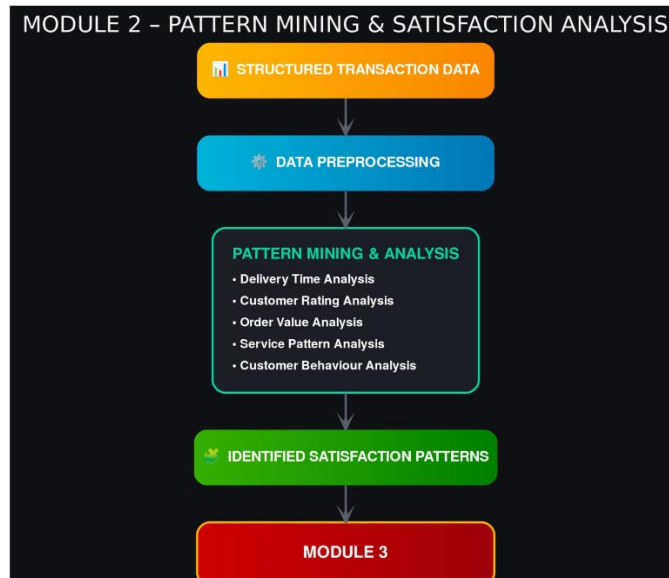
3.5 Selected Design & Detailed Design

Fig No 3.1:Module 1



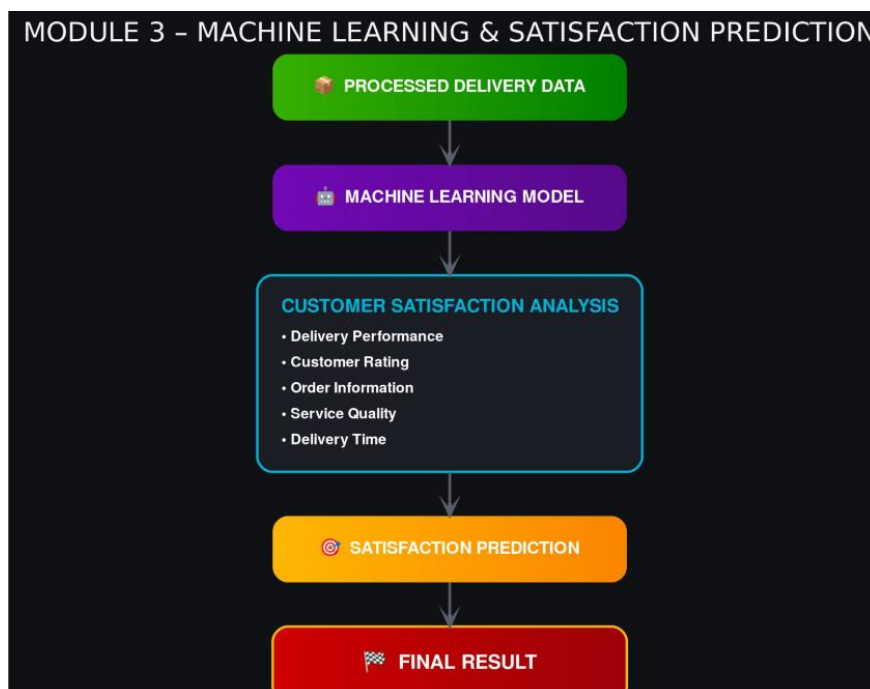
Satisfaction Prediction and Pattern Mining

Fig No 3.2:MODULE 2 – SATISFACTION PREDICTION AND PATTERN MINING



Customer Satisfaction Prediction

Fig No 3.3:MODULE 3 – CUSTOMER SATISFACTION PREDICTION



Satisfied / Neutral / Dissatisfied

System Interface

The system should provide clearly structured fields for:

- Order ID
- Customer ID
- Restaurant Information
- Order Value
- Delivery Time
- Delivery Distance
- Customer Rating
- Discount
- Payment Method
- Order Status

The system receives the transaction information and checks the data before storing it in the data warehouse.

3.6 Trade-off Analysis

Design Decision	Alternative	Benefit	Disadvantage	Final Decision
Data warehousing	Traditional data storage	Better organization	Requires structured design	Selected
Pattern mining	Manual analysis	Identifies hidden patterns	Requires data processing	Selected
Machine learning	Manual prediction	Faster prediction	Requires training data	Selected
Modular design	Single integrated program	Easier maintenance	Requires module integration	Selected
Multiple parameters	Single parameter	Better prediction information	More data required	Multiple parameters selected
Real-time analytics	Offline analysis	Faster insights	Requires processing resources	Selected

3.6 Sustainability, Ethics, Safety, Risk & Security

Sustainability

The project is primarily software-based, so it requires limited physical resources. A modular system architecture allows future improvements and additional features without completely redesigning the system.

Ethics

Customer information should be handled responsibly. The system should not unnecessarily expose personal customer information. Predictions should be used only for improving service analysis and customer experience.

Safety

Incorrect data or prediction results may lead to incorrect understanding of customer satisfaction. Therefore, the system should clearly present predictions as analytical results based on available data.

Security

Food delivery transaction information may contain customer-related data. The system should minimize unnecessary data exposure and ensure controlled data storage and handling.

Risk Register

Table 3.4

Risk	Likelihood	Severity	Risk Level	Mitigation
Incorrect transaction data	Medium	Medium	Medium	Data validation
Missing transaction data	Medium	Medium	Medium	Required-field checks
Inconsistent data	Medium	High	High	Data preprocessing
Data exposure	Low	High	Medium	Controlled data handling
Incorrect satisfaction prediction	Medium	High	High	Model evaluation
Large data volume	Medium	Medium	Medium	Efficient data warehousing
Poor data quality	Medium	High	High	Data cleaning and validation

CHAPTER 4

IMPLEMENTATION, TESTING & RESULTS, PROJECT MANAGEMENT

4.1 Development Approach & Resources

The development of the project follows a **modular approach** consisting of three modules.

Development Steps

1. Identify required food delivery transaction parameters.
2. Collect food delivery transaction data.
3. Design the data warehousing structure.
4. Implement data cleaning and organization.
5. Perform satisfaction pattern mining.
6. Prepare data for customer satisfaction prediction.
7. Implement the prediction model.
8. Test different transaction data conditions.
9. Generate customer satisfaction results.
10. Integrate all three modules.

Resources

Table 4.1

Resource	Purpose
Python	Application development
Python libraries	Data processing and analysis
Food delivery dataset	Analytics and prediction data
Computer	Development and testing
Data warehouse	Transaction data storage
Machine Learning tools	Customer satisfaction prediction

4.2 Hardware / Software / Fabrication Implementation & Modern Tools Used

Python

Python is used as the primary programming language for the project.

Data Warehousing

The food delivery transaction data is collected and organized into a structured format. The data warehouse stores information related to orders, customers, delivery, ratings, and other relevant transaction parameters.

Data Processing

Python-based processing is used to clean, organize, and prepare food delivery transaction data before pattern mining and machine-learning prediction.

Pattern Mining

Pattern mining techniques are used to analyze delivery transactions and identify relationships between delivery performance and customer satisfaction.

Machine Learning

Machine-learning techniques are used in the final module to predict customer satisfaction based on delivery-related parameters.

The prediction model analyses factors such as:

- Delivery time
- Order value
- Customer rating
- Discount
- Order information
- Delivery performance

4.3 Test Plan & Verification

Table 4.2

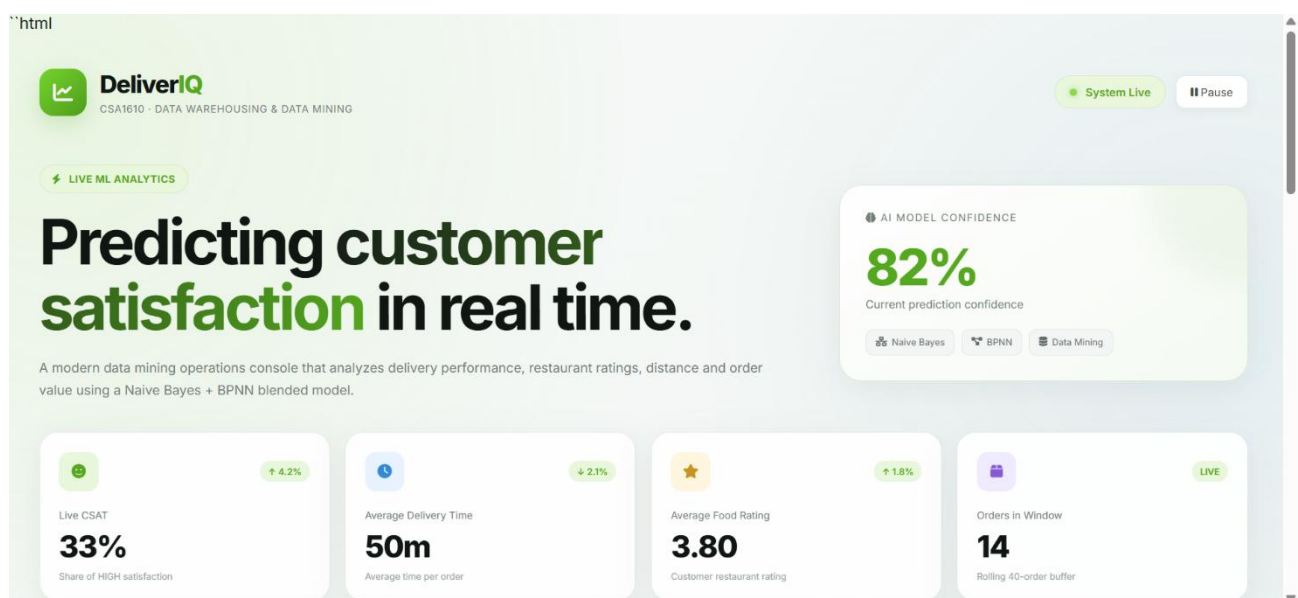
Requirement	Test Method	Acceptance Criterion
Transaction data entry	Enter valid transaction data	Values accepted
Missing field	Leave required field empty	System identifies missing data
Numerical input	Enter numerical values	Values accepted

Invalid input	Enter inappropriate format	System rejects or flags data
Data storage	Submit valid transaction data	Structured data stored
Pattern analysis	Process transaction data	Satisfaction patterns identified
Satisfaction prediction	Submit processed data	Prediction generated
Multiple parameters	Enter all required parameters	Complete data collected

4.4 Results

The expected output of the complete project is:

Fig No 4.1:Results



Customer Satisfaction Prediction Result

The project establishes a complete analytics pipeline by collecting food delivery transaction data, organizing the information, identifying customer satisfaction patterns, and generating a customer satisfaction prediction.

Fig No 4.2: System Workflow

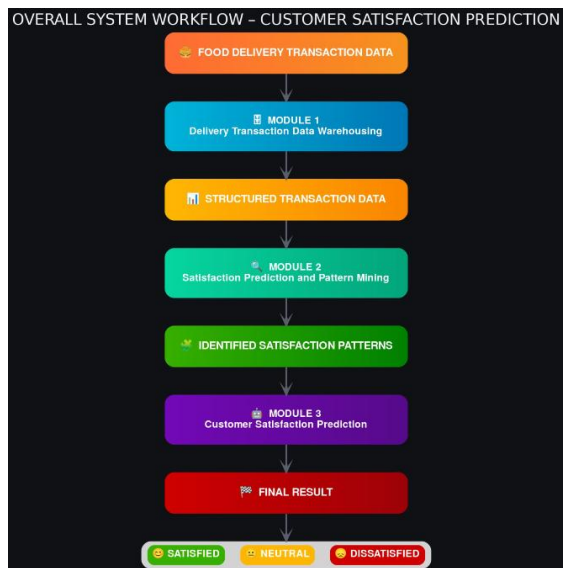
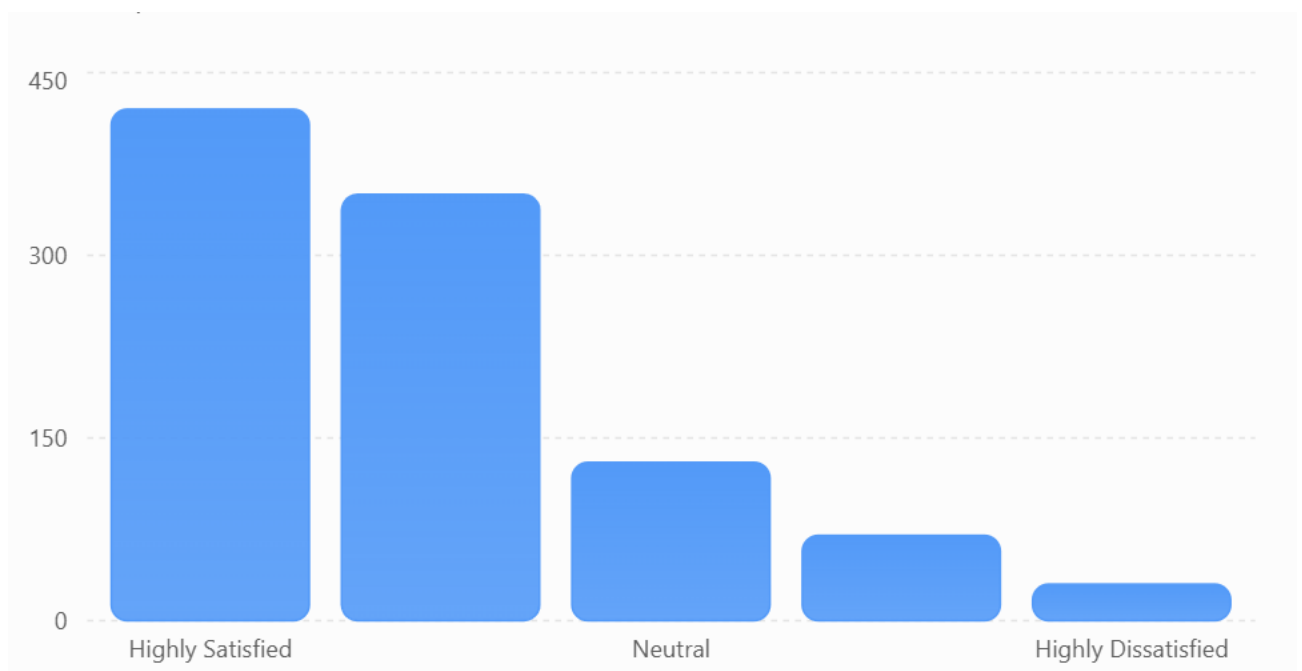


Fig No 4.3: Graph:



4.5 Data Analysis & Engineering Judgment

The quality of food delivery transaction data directly influences the quality of pattern analysis and customer satisfaction prediction.

A well-designed data warehousing and analytics system provides:

- Structured transaction information.

- Reduced data inconsistencies.
- Better data organization.
- Identification of important customer satisfaction patterns.
- Better compatibility with machine-learning processing.
- Easier integration between different modules.

The engineering decision to use a modular system allows each stage of the project to be independently developed and tested before complete system integration.

Module 1 focuses on transaction data organization.

Module 2 focuses on satisfaction pattern mining and analysis.

Module 3 focuses on customer satisfaction prediction.

4.6 Comparison with Existing Solutions & Achievement of Objectives

Table 4.3

Objective	Evidence	Achievement
Identify delivery transaction parameters	Order details, ratings, delivery time	Fully Achieved
Collect transaction information	Delivery Data Module	Fully Achieved
Organize transaction data	Data Warehousing Module	Fully Achieved
Analyze satisfaction patterns	Pattern Mining Module	Fully Achieved
Predict customer satisfaction	Machine Learning Model	Fully Achieved
Integrate all modules	Modular system architecture	Achieved
Provide scalable architecture	Modular design	Achieved

4.7 Strengths & Limitations

Strengths

- Structured delivery transaction data.
- Easy data organization.
- Supports large amounts of transaction information.
- Pattern mining capability.
- Customer satisfaction analysis.

- Modular architecture.
- Easy integration between modules.
- Can be extended with additional parameters.
- Supports future real-time analytics.

Limitations

- Depends on the quality of available transaction data.
- Missing or incorrect data can affect prediction.
- Customer satisfaction can depend on factors not available in the dataset.
- Real-time processing may require additional computing resources.
- Machine-learning prediction accuracy depends on training data.
- Customer data privacy and security must be carefully managed.

4.8 Project Planning, Roles & Budget

Project Timeline

Phase	Activity
Phase 1	Problem identification
Phase 2	Literature review
Phase 3	Identification of food delivery parameters
Phase 4	Delivery transaction data warehouse design
Phase 5	Data collection and preprocessing
Phase 6	Pattern mining and satisfaction analysis
Phase 7	Customer satisfaction prediction
Phase 8	Testing
Phase 9	Integration of all modules
Phase 10	Documentation

Project Roles

Module	Responsibility
Module 1	Delivery Transaction Data Warehousing
Module 2	Satisfaction Prediction and Pattern Mining
Module 3	Customer Satisfaction Prediction
Team	Testing, integration and documentation

Budget

Since the project is primarily software-based, the primary resources are an existing computer, Python software, development tools, and food delivery datasets.

Item	Estimated Cost
Computer/Laptop	Existing resource
Python	Free/Open-source
Food Delivery Dataset	Research/Educational dataset
Python Libraries	Free/Open-source
Development Tools	Free/Open-source
Data Storage	Minimal/Existing resource
Total Additional Cost	Minimal

Final Project Outcome

Food Delivery Data → Data Warehousing → Pattern Mining → Customer Satisfaction Analysis →
Machine Learning Prediction → Customer Satisfaction Result

CHAPTER 5

CONCLUSION, FUTURE WORK AND REFLECTION

5.1 Conclusion

The project Real-Time Food Delivery Analytics and Customer Satisfaction Prediction provides a structured approach for analysing food delivery transaction data and predicting customer satisfaction.

Module 1 – Delivery Transaction Data Warehousing provides the foundation for the complete system. This module collects, organizes, and stores important food delivery transaction information, including order details, delivery time, customer ratings, order value, discounts, restaurant information, and delivery status.

Module 2 – Satisfaction Prediction and Pattern Mining analyses the structured transaction data to identify important patterns and relationships related to customer satisfaction. Factors such as delivery performance, customer ratings, order value, discounts, and other order-related information can be analysed to understand customer behaviour and satisfaction trends.

Module 3 – Customer Satisfaction Prediction uses the processed and analysed information to predict customer satisfaction. The system can classify the expected satisfaction level based on available food delivery and customer-related data.

The quality of transaction data is important because it directly affects the reliability of pattern analysis and customer satisfaction prediction. By following a structured and modular approach, the proposed system establishes a clear connection between transaction data collection, data warehousing, data analysis, pattern mining, and machine-learning-based prediction.

Overall, the project combines delivery transaction data warehousing, satisfaction pattern mining, and customer satisfaction prediction into a unified analytics system. The application is intended to support food delivery analytics and provide better insights into the factors that influence customer satisfaction.

5.2 Future Work

The proposed project can be further enhanced in future versions by incorporating the following improvements:

1. Adding more food delivery parameters, such as weather conditions, traffic conditions, distance, delivery partner information, and peak-hour indicators.
2. Supporting larger and more diverse datasets from multiple food delivery platforms.
3. Improving data validation, cleaning, and preprocessing techniques to enhance data quality.
4. Integrating real-time delivery tracking to monitor order progress and delivery

performance.

5. Connecting the system to a secure cloud-based database for scalable data storage and processing.
6. Developing a user-friendly web or mobile application for accessing analytics and prediction results.
7. Using advanced machine-learning and deep-learning algorithms to improve customer satisfaction prediction accuracy.
8. Adding customer feedback, ratings, and review analysis using natural language processing techniques.
9. Integrating restaurant performance analysis to identify restaurants that consistently provide better or poorer customer experiences.
10. Developing real-time customer satisfaction dashboards with interactive charts and visualizations.
11. Providing automatic alerts when poor delivery performance or low customer satisfaction is detected.
12. Supporting personalized recommendations for restaurants, delivery partners, and service improvements based on customer behaviour.
13. Implementing continuous model training using newly collected transaction data to improve prediction performance over time.
14. Adding model evaluation and monitoring features to measure prediction accuracy and identify changes in customer behaviour.

5.3 Professional Learning & Individual Reflection

New Knowledge Acquired

During the development of this project, the following knowledge and understanding were acquired:

- Learned how to design and develop a modular data analytics application.
- Understood the importance of structured and reliable transaction data.
- Learned the basic concepts and applications of data warehousing.
- Understood the importance of data cleaning and preprocessing in analytics.
- Learned about pattern mining techniques and their applications.
- Understood how machine-learning techniques can be used for customer satisfaction prediction.
- Learned how different system modules can be integrated into a complete analytics

solution.

- Improved understanding of real-time data analytics and transaction processing.
- Learned how data analysis can be used to identify factors influencing customer behaviour.
- Improved understanding of the relationship between data quality, analysis, and prediction accuracy.

Engineering & Professional Skills Developed

The project helped in developing and improving the following engineering and professional skills:

- Problem identification and analysis.
- System design and modular development.
- Python programming.
- Data collection and organization.
- Data warehousing.
- Data cleaning and preprocessing.
- Pattern mining and data analysis.
- Machine-learning concepts and implementation.
- Testing and debugging.
- Data visualization and interpretation.
- Technical documentation.
- Project planning and management.
- Team collaboration and communication.
- Problem-solving and analytical thinking.
- Understanding of real-world data analytics applications.

Individual Reflection

Working on this project provided practical experience in applying data analytics and machine-learning concepts to a real-world problem. The development process helped in understanding how raw transaction data can be transformed into meaningful information through data preprocessing, warehousing, pattern analysis, and prediction.

The project also improved programming, analytical, problem-solving, testing, and documentation skills. Working with different modules helped develop a better understanding of how individual components can be integrated to form a complete software solution. Overall, the project provided valuable technical and professional learning that can be applied to future data analytics and machine-learning projects.

Individual Reflection

During the development of this project, one of the main engineering challenges was designing a structured system that could organize food delivery transaction data and connect it with satisfaction analysis and prediction.

The importance of data quality became clear because missing, incorrect or inconsistent transaction information can affect the later stages of pattern mining and customer satisfaction prediction.

The modular architecture was selected because it allows each part of the system to be developed and tested independently.

Module 1 focuses on collecting and organizing delivery transaction information through data warehousing.

Module 2 focuses on analyzing transaction data and identifying customer satisfaction patterns.

Module 3 focuses on using the processed information to predict customer satisfaction.

Through this project, knowledge of Python programming, structured data handling, data analytics, pattern mining and machine-learning workflows was improved.

The project also helped in understanding how raw transaction information can be converted into useful insights for analyzing customer behaviour and satisfaction.

If the project were redesigned or extended in the future, additional food delivery parameters, real-time data integration, advanced machine-learning techniques, customer feedback analysis and secure cloud-based data storage would be considered.

The overall project demonstrates how **Delivery Transaction Data Warehousing, Satisfaction Pattern Mining and Customer Satisfaction Prediction** can be integrated into a single system to support real-time food delivery analytics and better understanding of customer satisfaction.

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3. Kuo Ren Tan et al., “Evaluation of Machine Learning Methods Developed for Prediction of Diabetes Complications,” 2021.
4. Liangjun Jiang, “Diabetes risk prediction model based on community follow-up data using machine learning,” 2023.
5. Farida Mohsen, “A scoping review of artificial intelligence-based methods for diabetes risk prediction.”
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7. Relevant diabetes prediction dataset used for project development.

APPENDICES

APPENDIX A – DETAILED CALCULATIONS

For **Module 1 – Delivery Transaction Data Warehousing**, there are no major mathematical calculations because the module primarily focuses on collecting, validating, organizing, and storing food delivery transaction data.

However, basic transaction-related calculations can be documented in this appendix.

A.1 Order Value Calculation

The total order value can be calculated based on the item price, quantity, delivery charge, taxes, and applicable discounts.

Total Order Value = Item Cost + Delivery Charge + Tax – Discount

Example:

If:

- Item Cost = ₹500
- Delivery Charge = ₹40
- Tax = ₹30
- Discount = ₹70

Then:

Total Order Value = 500 + 40 + 30 – 70

Total Order Value = ₹500

The calculated order value can be stored as one of the transaction attributes.

A.2 Delivery Time Calculation

Delivery time can be calculated using the difference between the order placement time and the actual delivery time.

Delivery Time = Actual Delivery Time – Order Time

Example:

If:

- Order Time = 7:00 PM
- Delivery Time = 7:35 PM

Then:

Delivery Time = 35 minutes

This value can be used to analyse delivery performance and its relationship with customer satisfaction.

A.3 Discount Calculation

The discount amount can be calculated using:

Discount Amount = Original Order Value × Discount Percentage / 100

Example:

If:

- Original Order Value = ₹800
- Discount Percentage = 10%

Then:

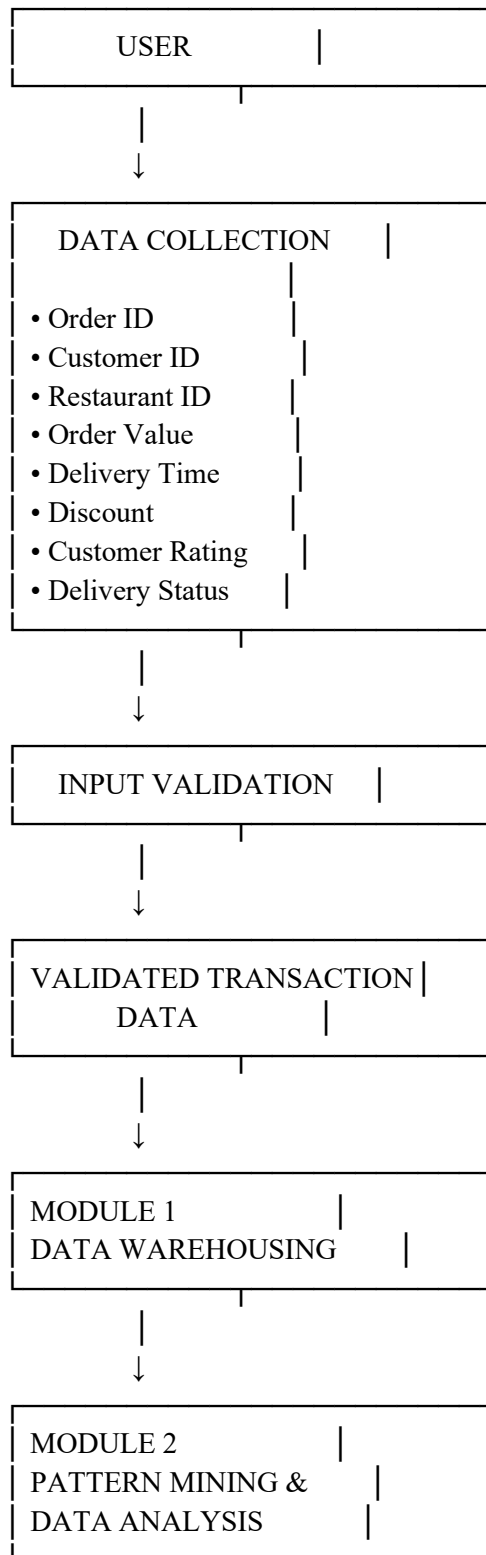
Discount Amount = 800 × 10 / 100

Discount Amount = ₹80

The discount information can be used during transaction analysis and customer behaviour analysis.

APPENDIX B – MODULE 1 PROCESS FLOW

B.1 Data Collection and Validation Flow



B.2 Module 1 Process

START



Open Application



Enter Transaction Details



Enter Delivery Information

```

↓
Enter Customer Rating
↓
Validate Input
↓
Is Input Valid?
↓
Yes → Store Transaction Data
↓
Prepare Data for Module 2
↓
END

```

APPENDIX C – SAMPLE DATA COLLECTION CODE

```

print("REAL-TIME FOOD DELIVERY ANALYTICS")
print("MODULE 1 - DATA COLLECTION")

order_id = input("Enter Order ID: ")
customer_id = input("Enter Customer ID: ")
restaurant_id = input("Enter Restaurant ID: ")

order_value = float(input("Enter Order Value: "))
delivery_time = float(input("Enter Delivery Time (minutes): "))
discount = float(input("Enter Discount Amount: "))
rating = float(input("Enter Customer Rating (1-5): "))

if not order_id:
    print("Invalid Order ID")

elif not customer_id:
    print("Invalid Customer ID")

elif not restaurant_id:
    print("Invalid Restaurant ID")

elif order_value <= 0:
    print("Invalid order value")

elif delivery_time <= 0:
    print("Invalid delivery time")

elif discount < 0:
    print("Invalid discount")

elif rating < 1 or rating > 5:
    print("Invalid customer rating")

else:
    print("\nTransaction data collected successfully")

```

```

print("Order ID:", order_id)
print("Customer ID:", customer_id)
print("Restaurant ID:", restaurant_id)
print("Order Value:", order_value)
print("Delivery Time:", delivery_time)
print("Discount:", discount)
print("Customer Rating:", rating)

print("\nValidated transaction data")

```

APPENDIX D – TRANSACTION DATA SPECIFICATION

D.1 Transaction Input Data Specification

S.No.	Parameter	Data Type	Purpose
1	Order ID	String	Uniquely identifies the food delivery order
2	Customer ID	String	Identifies the customer associated with the order
3	Restaurant ID	String	Identifies the restaurant associated with the order
4	Order Value	Float	Represents the total value of the order
5	Delivery Time	Float	Represents delivery duration in minutes
6	Discount	Float	Represents the discount applied to the order
7	Customer Rating	Float	Represents the customer's rating for the order
8	Delivery Status	String	Represents the current or final delivery status

D.2 Data Validation Specification

Parameter	Validation
Order ID	Must not be empty
Customer ID	Must not be empty
Restaurant ID	Must not be empty
Order Value	Numerical and positive
Delivery Time	Numerical and positive
Discount	Numerical and non-negative
Customer Rating	Numerical value between 1 and 5
Delivery Status	Must contain a valid status

APPENDIX E – SAMPLE TEST DATA

Test ID	Order Value (₹)	Delivery Time (min)	Discount (₹)	Rating	Input Status
T01	450	30	50	5	Valid
T02	750	42	75	4	Valid
T03	320	25	20	5	Valid
T04	—	35	40	4	Invalid
T05	600	—	30	3	Invalid
T06	500	40	25	—	Invalid
T07	-250	30	20	4	Invalid
T08	800	55	100	6	Invalid

E.1 Purpose of Test Data

The sample data is used to verify whether:

- Complete transaction data is accepted.
- Missing transaction values are identified.
- Numerical input is processed correctly.
- Negative or invalid values are rejected.

- Customer ratings outside the valid range are rejected.
- Validated transaction data is prepared for Module 2.
- Data validation works correctly before storage and analysis.

Note: These are sample test values for demonstrating the appendix and should be replaced with actual project test data where applicable.

APPENDIX F – ETHICAL CONSIDERATIONS

The **Real-Time Food Delivery Analytics and Customer Satisfaction Prediction** system may process customer, restaurant, transaction, and feedback-related information. Therefore, responsible data handling is important.

The following ethical principles should be followed:

1. Customer information should be collected only for the intended project purpose.
2. Personally identifiable information should not be unnecessarily stored.
3. Sensitive customer information should be protected from unauthorized access.
4. Transaction data should not be shared without appropriate authorization.
5. Customer ratings and feedback should be analysed fairly and without manipulation.
6. The prediction results should be presented as analytical predictions rather than guaranteed outcomes.
7. The system should avoid unfairly discriminating against customers, restaurants, or delivery partners.
8. Data should be anonymized or appropriately protected wherever possible.
9. Users should be informed about how their data is being used when applicable.
10. Appropriate security measures should be implemented when storing or transmitting transaction data.

F.1 Institutional Approval

If institutional approval is required by the department, the following information may be included:

- Approval document
- Approval number
- Date of approval
- Faculty or department authorization

If institutional approval is not applicable, the following statement may be used:

Ethical/institutional approval was not applicable to this software-based academic project because the project does not involve clinical intervention or direct medical treatment.

APPENDIX G – PROJECT PROGRESS RECORD

Meeting	Date	Discussion	Work Completed	Next Activity
1	_____	Project topic selection	Topic finalized	Literature review
2	_____	Project requirements	Requirements identified	Module design
3	_____	Module 1 discussion	Transaction parameters identified	Data collection implementation
4	_____	Python implementation	Initial code developed	Input validation
5	_____	Data validation and testing	Test cases prepared	Error correction
6	_____	Module integration	Module 1 connected to Module 2	Final testing
7	_____	Report preparation	Documentation completed	Final submission

G.1 Meeting Discussion Points

During project meetings, the following points can be recorded:

- Selection of food delivery analytics as the project topic.
- Identification of relevant transaction parameters.
- Design of the Delivery Transaction Data Warehousing module.
- Selection of Python and other required technologies.

- Data validation and preprocessing requirements.
- Design of satisfaction pattern mining.
- Development of the customer satisfaction prediction module.
- Integration of the project modules.
- Testing and debugging.
- Report preparation and final documentation.

APPENDIX H – SAMPLE DATASET STRUCTURE

Order ID	Customer ID	Restaurant ID	Order Value	Delivery Time	Discount	Rating	Delivery Status
ORD001	C001	R101	450	30	50	5	Delivered
ORD002	C002	R102	750	42	75	4	Delivered
ORD003	C003	R103	320	25	20	5	Delivered
ORD004	C004	R101	600	55	30	3	Delivered
ORD005	C005	R104	850	65	100	2	Delivered

This sample structure demonstrates how food delivery transaction information can be organized before being processed by the analytics and prediction modules.

APPENDIX I – FEEDBACK FORM

I.1 User Feedback Form

S.No.	Evaluation Area	Rating / Feedback
1	Ease of data entry	_____
2	Clarity of transaction parameters	_____
3	Ease of understanding	_____
4	Input validation	_____
5	Clarity of prediction results	_____
6	Overall usability	_____
7	Suggestions for improvement	_____

I.2 Sample Feedback Questions

1. Is the transaction data-entry process easy to understand?
2. Are the required food delivery parameters clearly displayed?
3. Is the input validation understandable?
4. Is the application easy to use?
5. Are the customer satisfaction results easy to understand?
6. Are the analytics and visualizations useful?
7. What additional parameters should be included?
8. What improvements would make the application more useful?

APPENDIX J – INDIVIDUAL CONTRIBUTION EVIDENCE

Student	Assigned Responsibility	Design & Development	Testing & Analysis	Report Contribution	Contribution %
Student 1	Module 1 – Data Warehousing	Transaction data collection and storage	Module testing	Chapter 1, Chapter 4	____%
Student 2	Module 2 – Pattern Mining & Analysis	Data preprocessing and pattern mining	Pattern analysis	Chapter 2, Chapter 3	____%
Student 3	Module 3 – Satisfaction	Prediction model and application	Prediction testing and	Chapter 4, Chapter 5	____%

	Prediction	integration	result analysis		
Student 4	Documentation & Testing	Interface and documentation	System testing	References and Appendices	____ %
Total					100%

MODULE 1 – INDIVIDUAL CONTRIBUTION

If you are specifically responsible for **Module 1 – Delivery Transaction Data Warehousing**, your contribution can be described as:

- Identified the required food delivery transaction parameters.
- Designed the Delivery Transaction Data Warehousing module.
- Implemented transaction data collection.
- Implemented input validation.
- Organized transaction data into a structured format.
- Tested valid and invalid transaction inputs.
- Prepared transaction data for further analysis.
- Assisted in integrating Module 1 with Module 2.
- Supported data preprocessing and data quality checking.
- Prepared technical documentation related to Module 1.

CAPSTONE PROJECT OUTCOME MAPPING SHEET

To be completed by the department/faculty assessor.

Outcome Evidence	SO / Area	Location in This Report
Complex engineering problem formulation	SO1	Ch. 1, Ch. 2
Engineering design under constraints	SO2	Ch. 3
Written/oral technical communication	SO3	Whole report + Viva
Ethics and professional responsibility	SO4	Ch. 3 (3.7)
Teamwork and leadership	SO5	Ch. 4 (4.8)
Experimentation, analysis, engineering judgment	SO6	Ch. 4 (4.3–4.6)
Independent acquisition of new knowledge	SO7	Ch. 5 (5.3)
Standards	SO2	Ch. 3 (3.2)
Sustainability and societal/environmental impact	SO2/SO4	Ch. 3 (3.7)
Risk assessment and mitigation	SO2/SO4	Ch. 3 (3.7)
Security considerations	Relevant SO	Ch. 3 (3.7)
Integrated/system-level engineering	SO1/SO2	Ch. 3 (3.5)
Project planning and management	SO5	Ch. 4 (4.8)
Individual contribution	SO1–SO7	Contribution Statement + Ch. 4 (4.8)